

Phish Guard: A Deep Learning Framework for Real-Time URL-Based Phishing Detection

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Abstract— *The prevalence of phishing attacks, where malicious features attempt to steal sensitive information by impersonating legitimate websites, poses a several threat to online security. To combat this threat, this project introduces Fresh-Phish, an open-source and extendable system designed to detect phishing websites using machine learning and ANN algorithms. Unlike existing approaches, Fresh-Phish addresses key drawbacks by providing a framework for extracting features and maintaining an up-to-date dataset of phishing and genuine websites. By integrating 29 distinct characteristics, Fresh-Phish aims to accurately differentiate between authentic and phishing websites, enhancing the ability to protect internet users from malicious attacks. Model approach improves upon existing methods by addressing three critical issues. Firstly, it provides a systematic framework for feature extraction and dataset management, ensuring the dataset remains current and comprehensive. Secondly, it reduces the reliance on a vast number of features by selecting 29 key characteristics, supported by evidence of their relevance in training the machine learning and ANN classifier. Finally, Fresh-Phish aims to mitigate dataset bias by ensuring a balanced representation of URL and content-based attributes, improving the classifier's ability to generalize and detect phishing websites effectively.*

Keywords— Artificial Intelligence, Phishing, Deep Learning, ANN, URL.

INTRODUCTION

The intimate has revolutionized the way we live, work, and communicate, offering unprecedented convenience and connectivity. However, this digital landscape is not without its dangers. One of the most pervasive threats facing internet users today is phishing, a form of cyberattack where malicious actors impersonate legitimate entities to deceive users into providing sensitive information such as usernames, passwords, and credit card details. Phishing attacks have become increasingly common and sophisticated, putting internet

users at risk of identity theft, financial loss, and other forms of cybercrime.

Detecting phishing websites is a challenging task due to the evolving nature of these attacks. While various countermeasures have been proposed by academia, business, and research groups, phishing attacks continue to pose a threat. Machine learning and ANN algorithms offer a promising approach to discriminating between phishing and authentic websites, leveraging patterns in website characteristics to identify malicious intent. However, existing machine learning and ANN algorithms for phishing detection have several limitations that hinder their effectiveness.

The first major drawback is the lack of a comprehensive framework for extracting features and maintaining an up-to-date dataset of phishing and genuine websites. This results in outdated and incomplete datasets, leading to reduced accuracy in phishing detection. Secondly, existing algorithms often employ a vast number of features, with limited evidence supporting the relevance of these features in distinguishing between phishing and authentic websites. This not only increases computational complexity but also makes it challenging to interpret the results. Finally, existing datasets used in research are often skewed, with an imbalance in URL or content-based attributes, which can lead to biased classifiers that fail to generalize well to new data.

WHAT IS A PHISHING?

Phishing is a type of digital theft that disguises itself as legitimate or genuine sources to steal users' private and confidential information. It has become a popular attacking approach in cyberspace by utilizing web applications' vulnerabilities and end users' ignorance, which is a security issue that needs to be addressed.

The evolution of phishing attacks is illustrated in Figure 1. Back in 1996, the term "phishing" was first

introduced, and phishing attacks were slowly spread through various communication media over the years.



Figure 1: Evolution of phishing attacks from 1996 to 2020.

It started with spam messages, mobile malware, spear-phishing to "Man in the Middle", Vishing, "Chat in the Middle", "Tabnabbing", ¹ "Xbox Live", etc. Phishing attacks started becoming a serious issue and caught more attention among researchers when a major incident happened in 2014, causing a huge financial loss. With the advent of the Internet and the popularity of social media, the number of phishing attacks has increased rapidly since 2016 and continued to grow in an upward trend. According to the latest statistics from APWG (Anti Phishing Working Group), the number of phishing attacks has grown tremendously since March 2020 and doubled over the course of the year.

Since phishing has become a serious security issue, understanding how it operates is an utmost important task in the detection and prevention of such cybersecurity threat. The life cycle of a typical phishing attack is shown in Figure 2, consisting of five phases.



Figure 2: Phishing attack life cycle.

The first phase is called reconnaissance or planning phase, in which the phishers choose the communication media, select the phishing vector, and identify potential victims. The second phase is weaponization or preparation phase, whereby phishers prepare phishing materials to be propagated to their targeted victims. The next stage is distribution or phishing phase, as phishers start to deploy the baits by delivering the phishing materials to victims.

The following stage is called exploitation or penetration phase, where phishers exploit victims' weaknesses by luring them into giving up their private and confidential information. The final stage is known as exfiltration or data acquisition phase. The phishing operation has succeeded at this point, and phishers had successfully obtained the information they intended to take when planning the phishing attack initially. Phishers can decide to take further actions to gain financial benefits, or use the collected information for other purposes.

Deep Learning(DL)

Phishing appears to be an effective way for cybercrime to occur because most users are unable to identify phishing websites or emails. One of the current challenges in dealing with cyberthreats, especially phishing attacks, is lacking of cyber security solution, and Artificial Intelligence (AI) is believed to be the next frontier in cyber security defense.

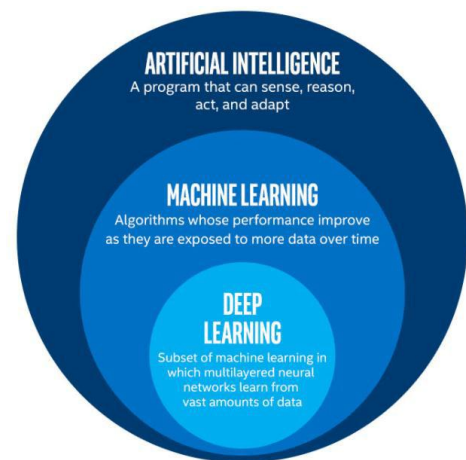


Figure 3: Venn diagram of AI, ML and DL

ML is a part of AI that teaches machine the ability to learn like human beings. DL is a subset of ML derived from a neural network model (FIGURE 3). Traditional ML techniques refer to the learning methods that require human expertise to perform feature extraction and selection. Feature selection is separated from classification task in a classical ML model, and these two processes cannot be combined together to optimize the model's performance. However, DL fills this gap by integrating these two processes in a single phase to detect and classify phishing attacks effectively and efficiently. Although traditional ML approaches provide high accuracy and low false-positive rate, they still require manual feature engineering and depend on third-party services. In contrast, DL models can learn and extract features automatically without human intervention. This eliminates the need for manual feature engineering and third-party services dependency. Moreover, traditional ML with manual feature engineering fails to deal with multi-dimensional and large-scale datasets in the big data era. DL, however, can handle a significant amount of data and becomes a powerful tool for phishing detection that requires more attention in the cybersecurity community. There was no study that combined DL and phishing detection in a SLR approach despite the increasing attention given to these two domains. Therefore, a detailed process of selecting relevant studies was described in this paper, to examine the

current trends and patterns in the existing research on DL for phishing detection. The primary purpose of conducting this SLR is to analyze the pros and cons of the state-of-the-art DL techniques, identify the current issues, highlight the research gaps, and recommend future research directions.

System Design

User Interface Module: The User Interface Module serves as the entry point for users to interact with the Fresh-Phish system. It provides a user-friendly web interface for uploading URLs for analysis, viewing analysis results, and providing feedback. This module includes components such as forms for uploading URLs (uploadURL()), result display pages for viewing analysis results (viewAnalysisResults()), and feedback forms for users to provide additional information (provideFeedback()). The User Interface Module relies on the Feature Extraction Module to process uploaded URLs.

Feature Extraction Module: The Feature Extraction Module is responsible for extracting relevant features from URLs to be used for phishing detection. This module includes components for extracting URL-based features (e.g., domain age, URL length), content-based features (e.g., presence of keywords), and server-based features (e.g., server location, SSL certificate). The Feature Extraction Module depends on the User Interface Module to receive URLs for analysis and the Machine learning and ANN Module to process the extracted features.

ANN Module: This module includes components for training the classifier (trainClassifier()), predicting the class of URLs (predictClass()), and updating the dataset (updateDataset()). The Machine learning and ANN Module depends on the Feature Extrusion for Finding extracted features and the Dataset Management Module for imaging the dataset used for training.

Dataset Management Module: The Dataset Management Module manages the dataset used for training the machine learning and ANN classifier. This module includes components for updating the dataset with new URLs (updateDataset()), verifying the authenticity of websites (verifyWebsite()), and maintaining the dataset (maintain Dataset()). The Dataset Management Module depends on the Machine learning and ANN Module for updating the dataset and the Feature Extraction Module for verifying website authenticity.

Mechanism

Training Phase: First, the system provides users the ability to upload URLs for analysis via the user interface. The Feature Extraction Module processes these URLs and extracts 30 features from every URL. The features extracted are URL length, domain age, keywords present, server location, and SSL certificate details. The features are then utilized to train the ANN and machine learning classifier in the Machine learning and ANN Module. The classifier is trained using a dataset containing safe and phishing sites, with the proportion of safe to phishing sites being considered.

Testing Phase: During the testing phase, when a user enters a URL to analyze, the system employs BeautifulSoup4 to extract the same 30 features from the

URL. These features are then reshaped into a 1 x 30 array and input into the trained classifier. The classifier returns the probability of the URL being phishing based on its features.

Data Flow:

The Data Flow Diagram (DFD) depicts the flow of information between different parts of the system. The figure 4 illustrates how the system is accessed by users who upload URLs to be analyzed. This information is processed by the Feature Extraction Module.

Which extracts the appropriate features from the URLs. These characteristics are subsequently transferred to the Machine learning and ANN Module, where they are utilized for training the machine learning and ANN classifier. The trained classifier can predict whether a URL is secure or phishing from its characteristics. The Machine learning and ANN Module also communicates with the Dataset Management Module, whose task is to update the dataset utilized for classifier training. This helps to ensure that the classifier is accurate and current in its capacity to separate safe and phishing URLs. Further, if a URL is detected as phishing, the Machine learning and ANN Module triggers an alert via the Telegram bot, which informs the user of the potential threat. This alert system is important to make sure that the users are informed in a timely manner about phishing attempts and take necessary action to safeguard themselves.

In general, the DFD gives a general idea of data flow in the Fresh Phish system, with emphasis on major components and how they interact with each other. The diagram illustrates how data flows from entities and processes within the system, from user input to feature extraction, machine learning and ANN prediction, dataset management, and alerting. This vivid visualization assists in understanding the system functionality and how the various components cooperate to identify phishing websites and secure users against online threats.

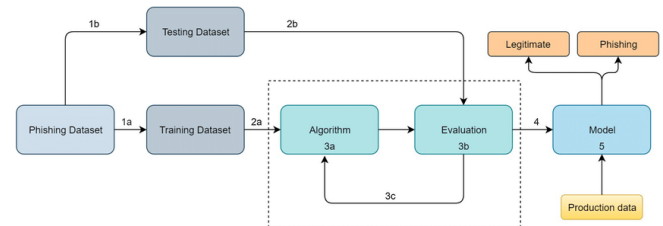


Figure 4: Data Flow Diagram

Interface Diagram

The interface diagram as shown in Figure 5 for the Fresh-Phish project illustrates the interaction between various components of the system, focusing on the interfaces exposed by each component.

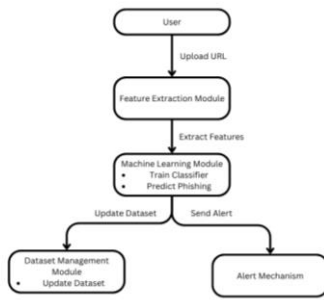


Figure 5: Interface Diagram

1. **User Interface:** This interface represents the user-facing aspects of the system. It includes methods for uploading URLs for analysis (uploadURL()), viewing analysis results (viewAnalysisResults()), and providing feedback (provideFeedback()). Users interact with the system through this interface, initiating the analysis process and receiving information about the analysis results.
2. **Interaction Flow:** The diagram shows the flow of interactions between the components. The User interface component uploads a URL, which is then processed by the Feature Extraction Module to extract relevant features. These features are passed to the Machine Learning Module for analysis, which determines if the URL is safe or phishing. If the URL is classified as phishing, the Machine Learning Module sends an alert message.

Conclusion

In summary, the project is a great leap in the area of phishing detection and prevention. Utilizing machine learning and ANN algorithms and advanced feature extraction methods, the system can effectively distinguish between secure and phishing websites. The creation of an open-source and extensible system tackles the rising menace of phishing attacks and offers a useful tool for researchers and practitioners alike in the field of cybersecurity. The modular structure of the project, comprising modules like the User Interface Module, Feature Extraction Module, Machine learning and ANN Module, Dataset Management Module, and Telegram Bot Module, reflects a structured and holistic approach to phishing detection. The combination of these modules enables effective processing of URLs, feature extraction, and website classification, ultimately resulting in timely alert notifications to users.

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